A proposed density-one lower bound for strict Erdős–Straus solution counts

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Draft status. This manuscript presents a proof for specialist scrutiny. The prime exceptional-set lower bound and relative-density conclusion now have an end-to-end Lean certificate through a restricted-subfamily argument. The integer assertion, the manuscript's exact Proposition 3 route, the published-upper-bound integration used for the exponent-matching corollary, and the sharper Section 10 refinement are not all formalized in that certificate. The work has not received independent human specialist review, and literature-wide originality remains unestablished. This is not a proof of the Erdős–Straus conjecture.

Abstract

Let $R(n)$ count triples of positive integers $x < y < z$ satisfying $4/n = 1/x + 1/y + 1/z$. For a prime p, let $R_{\mathrm{II}}(p)$ count those triples with $p \mid y, z$ and $(p, x) = 1$. We present an argument for the existence of an absolute constant $c > 0$ such that

$$\#\left\{p \le N : p \text{ prime},\ p \ge n_0, R_{\mathrm{II}}(p) < c\frac{(\log p)^3}{\log\log p}\right\} \ll \frac{N(\log\log N)^3}{(\log N)^3}.$$

Here n_0 is an absolute fixed threshold. The same exceptional-set estimate holds for integers with $R(n)$ in place of $R_{\mathrm{II}}(p)$. The arithmetic step is a weighted compatible-pair estimate for rough, possibly composite witness moduli. An explicit second-moment argument transfers this estimate to finite intervals, and an injective decoder gives strictly ordered solutions. Combined with the published prime first-moment upper bounds of Elsholtz and Tao, the proposed lower bound gives $R_{\mathrm{II}}(p) = (\log p)^{3+o(1)}$ and $R(p) = (\log p)^{3+o(1)}$ on one set of primes of relative density one. If valid, this improves the density-one prime lower bound in Elsholtz–Tao's Theorem 1.8, replacing its logarithmic exponent $(\log 3)/2$ by $3 - o(1)$, already for strict Type-II solutions. A separate proposed refinement to $\Delta = O(\mu)$ reduces the exception bound to $O(N \log\log N/(\log N)^3)$; the main proof does not depend on this refinement. Neither result gives a leading constant, control every prime, or improve the classical zero-solution exceptional-set benchmark. The arithmetic estimates and their applications are separated to make the proof's dependencies explicit.

The prime lower-bound and relative-density assertion have also been proved in Lean by an alternative argument using a genuine restricted subfamily and weaker sufficient intermediate estimates. That certificate does not by itself formalize the all-integer assertion or the published upper-bound input used to identify the matching exponent as exactly three.

1. Statements and scope

All logarithms are natural. Constants in $\ll$ and $\asymp$ are absolute unless a fixed divisor-function index is specified. All asymptotic assertions concern sufficiently large arguments. Write τ_j for the j-fold divisor function, $\tau = \tau_2$, ω for the number of distinct prime factors, φ for Euler's function, and $P^-(m)$ for the least prime factor, with $P^-(1) = \infty$.

Define

$$R(n) = \#\{(x, y, z) \in \mathbb{N}^3 : 1 \le x < y < z, 4/n = 1/x + 1/y + 1/z\}.$$

For prime p, $R_{\mathrm{II}}(p)$ imposes the additional conditions $p \mid y$, $p \mid z$, and $(p, x) = 1$. The subscript is tied to these positions in the strictly ordered triple.

Theorem 1 (proposed abundance bound). There exist absolute constants $c, C > 0$ and $n_0 > e^e$ such that, for all sufficiently large N,

$$\#\left\{ n_0 \le n \le N : R(n) < c\frac{(\log n)^3}{\log\log n} \right\} \le C\frac{N(\log\log N)^3}{(\log N)^3}. \tag{1}$$

The same estimate holds when the set is replaced by

$$\left\{ n_0 \le p \le N : p \text{ prime},\ R_{\mathrm{II}}(p) < c\frac{(\log p)^3}{\log\log p} \right\}. \tag{2}$$

Corollary 2 (proposed typical prime exponent). There exists a set $\mathcal{P}_0$ of primes such that

$$\frac{\#\{p \le N : p \in \mathcal{P}_0\}}{\pi(N)} \longrightarrow 1$$

and, as $p \to \infty$ through this one set,

$$\frac{\log R_{\mathrm{II}}(p)}{\log\log p} \longrightarrow 3, \qquad \frac{\log R(p)}{\log\log p} \longrightarrow 3. \tag{3}$$

Theorem 1 is deduced below from Proposition 3. Its deduction is independent of how Proposition 3 is proved. Sections 5–9 give the proposed arithmetic proof, using the specified classical analytic inputs. A reader questioning one of those estimates can therefore still assess the conditional implication separately. Section 10 presents a stronger overlap claim and its sharper exception bound, with their additional argument and distinct review status. Sections 2–9 and Corollary 2 do not depend on Section 10.

The exceptional proportion in (2) is $O((\log\log N)^3/(\log N)^2)$ by the prime number theorem. It tends to zero, but its counting bound tends to infinity. No assertion about the remaining primes follows. In particular, (3) is not an asymptotic formula $R(p) \sim C_0(\log p)^3$ and does not settle the conjecture.

1.1. Relation to earlier work

Elsholtz–Tao's Theorem 1.8 gives $f(p) \ge (\log p)^{(\log 3)/2 - o(1)}$ on a set of primes of relative density one [ET, p. 56]. Theorem 1, if valid, improves this particular lower-bound conclusion: it gives $f(p) \ge R_{\mathrm{II}}(p) \ge c(\log p)^3/\log\log p$ on a set of primes of relative density one, with an explicit exceptional-set estimate. Thus the proposed lower logarithmic exponent increases from $(\log 3)/2 \approx$

0.5493 to $3-o(1)$, even for the strict Type-II subset. This comparison needs inclusion, not equality, of counting conventions.

This is not an improvement of every assertion in Theorem 1.8, nor of Elsholtz–Tao's first-moment bounds in Theorem 1.1. Their upper bounds supply the upper side of (3) and are credited as an input in Section 4. Improving this identified 2013 conclusion, if the proof is correct, is a precise comparison; being the first work to do so is a separate priority claim that this draft does not make.

Elsholtz–Planitzer study related lower bounds with different quantifiers [EP]. A statement on a density-one set of integers cannot, by itself, be restricted to primes. Dahan studies closely related witness systems and fixed and growing search depths [D, Sections 4.1–4.4]. These comparisons must include consequences of prime-witness mass estimates, not just the older printed exponent. This draft does not assert that all such consequences or subsequent literature have been exhausted.

The candidate contribution here is the compatible-pair estimate and its quantitative multiplicity consequence, not a new parametrization, CRT identity, second-moment principle, or published upper bound. The classical exceptional-set problem for *zero* solutions is different from (1)–(2).

2. Witness family and the arithmetic proposition

For $Y \to \infty$, put

$$X = 3Y, \qquad L = \log Y, \qquad w = L^{100}.$$

Retain all rows

$$M = 3u \le X, \qquad Q = 4M-1 = 12u-1, \qquad P^{-}(Q) > w.$$

Both prime and composite Q are included. Define

$$S_M = \{s \in \mathbb{N} : 1 \le s < M,\ s \mid M^2\}, \qquad \mathcal{A}_Y = \{(Q,s) : Q \text{ retained},\ s \in S_M\},$$

$$\mu = \sum_{(Q,s)\in\mathcal{A}_Y} \frac{1}{Q}.$$

Each label has the unique squarefree representation

$$s = rb^2, \qquad M = rbc, \qquad r \text{ squarefree}, \quad b < c.$$

Set $\kappa(s) = rb$ and $h_s = \operatorname{lcm}(3, \kappa(s))$. For fixed s, the divisibility and row conditions, apart from $s < M$ and roughness, are exactly $h_s \mid M$. Every label is coprime to its row modulus.

Define the unordered distinct-row overlap

$$\Delta = \sum_{\substack{Q<Q' \text{ retained} \\ g=(Q,Q')>1}} \frac{g}{QQ'} \#\{(s,t) \in S_M \times S_{M'} : s \equiv t \pmod{g}\}. \tag{4}$$

All moduli and divisor sums retain full prime powers.

Proposition 3 (arithmetic estimate claimed here). For this family,

$$\mu \asymp \frac{L^3}{\log w}, \qquad \Delta \ll L^3 \log(2L). \tag{5}$$

The stronger assertion $\Delta = O(\mu)$ is proposed separately in Section 10; it is not an assumption of the main argument in Sections 2–9.

3. From arithmetic estimates to interval abundance

3.1. An injective strict decoder

Lemma 4. Suppose $n > Q$, $s \in S_M$, and $Q \mid n + 4s$. Then

$$x = \frac{nM + s}{Q}, \qquad y = nM, \qquad z = \frac{nMx}{s} \tag{6}$$

is a positive integer solution counted by $R(n)$. If $n = p$ is prime, it is counted by $R_{\mathrm{II}}(p)$. For fixed n the map is injective.

Proof. Since $4(nM+s) = n(Q+1)+4s$ and $(4, Q) = 1$, x is integral. The equality $QMx = nM^2+Ms$, together with $s \mid M^2$ and $(s, Q) = 1$, implies $s \mid Mx$. Thus z is integral and divisible by n. The inequality $n > Q$ gives $x > M > s$, whence $z > y$. Also

$$x < \frac{(n + 1)M}{Q} \leq \frac{n + 1}{3} < n \leq y,$$

since $Q = 4M - 1 \geq 3M$ and $n > 2$. Finally $Qx = nM + s$ gives $4Mx = nM + x + s$, which is the desired reciprocal identity after division by nMx. For prime p, $1 \leq x < p$ gives $(p, x) = 1$. The inverse is

$$M = y/n, \qquad Q = 4M - 1, \qquad s = Qx - nM. \tag{7}$$

The triple is already strictly sorted, so this also rules out collisions after sorting. □

3.2. Complete-period variance and the interval endpoint

Put

$$T_Y(n) = \sum_{(Q,s)\in\mathcal{A}_Y} \mathbf{1}_{Q|n+4s}.$$

On the uniform probability space modulo the least common multiple of all retained Q, its mean is μ. Two indicators on different rows have intersection probability $g/(QQ')$ when $s \equiv t \pmod g$, and zero otherwise. For coprime rows the covariance is zero. Different labels on one row are disjoint, since $s, t < M < Q$ and $(4, Q) = 1$. Dropping negative covariances therefore gives

$$v := \mathrm{Var}_{\mathrm{CRT}}(T_Y) \leq \mu + 2\Delta. \tag{8}$$

Let $m = |\mathcal{A}_Y|$. Since $Q < 12Y$, $m/\mu \leq 12Y$. The sum of absolute formal coefficients of $F = 1 - T_Y/\mu$ is at most $1 + 12Y$. Each monomial in its square is either identically zero or the indicator of one residue class modulo its full least common multiple. On any interval J of H consecutive integers, such a class has counting error at most one. Expanding the square and bounding the errors in absolute value yields

$$\sum_{n\in J} \left(1 - \frac{T_Y(n)}{\mu}\right)^2 \leq H\frac{v}{\mu^2} + (1 + 12Y)^2. \tag{9}$$

Consequently

$$\#\{n \in J : |T_Y(n) - \mu| \geq \mu/2\} \leq 4\left(H\frac{v}{\mu^2} + (1 + 12Y)^2\right). \tag{10}$$

This retains the entire interval error; no equidistribution over a period longer than J has been assumed.

3.3. Proof of Theorem 1 from Proposition 3

By (5) and (8), $v/\mu^2 \ll (\log L)^3/L^3$. On $J = (N/2, N] \cap \mathbb{Z}$, choose $Y = N^{1/3}$. For sufficiently large N all $n \in J$ exceed $12Y$, and (10) implies, apart from

$$O\left(\frac{N(\log\log N)^3}{(\log N)^3} + N^{2/3}\right)$$

integers, that $T_Y(n) \geq \mu/2 \gg (\log N)^3/\log\log N$. Lemma 4 gives $R(n) \geq T_Y(n)$ and $R_{\mathrm{II}}(p) \geq T_Y(p)$. Choose a single sufficiently small absolute c for all large dyadic blocks.

To sum to N, discard the integers below $\sqrt{N}$. In the remaining dyadic blocks logarithms are comparable to $\log N$; the block lengths sum to $O(N)$ and the endpoint errors sum to $O(N^{2/3})$. Both the discarded prefix and the endpoints are absorbed in (1). The prime assertion follows from the same blockwise argument: any prime failing the Type-II threshold must be among the integers failing the packet threshold. This is stronger than merely restricting a density-one statement about $R(n)$. $\square$

4. The published upper side and proof of Corollary 2

With the notation of Elsholtz–Tao [ET, Theorem 1.1 and equation (1.4), pp. 53–54],

$$\sum_{p\leq N} f_{\mathrm{II}}(p) \ll N(\log N)^2, \qquad \sum_{p\leq N} f(p) \ll N(\log N)^2 \log\log N. \tag{11}$$

Their unrestricted solution count includes every strictly ordered triple here. Their Type-II count fixes the same divisibility positions and includes ours. Thus

$$R_{\mathrm{II}}(p) \leq f_{\mathrm{II}}(p), \qquad R(p) \leq f(p).$$

There is no use of $R(p) \leq f_{\mathrm{II}}(p)$ or an equality of conventions.

Markov's inequality on dyadic blocks, followed by summation above $\sqrt{N}$, shows that the primes violating $R(p) \leq C_1(\log p)^3(\log\log p)^2$ have relative density zero. This upper-tail consequence is already contained in [ET, p. 54], not a new result. Intersect its complement with the complement of the exceptional set (2). The resulting single set $\mathscr{P}_0$ has relative density one and satisfies

$$c\frac{(\log p)^3}{\log\log p} \leq R_{\mathrm{II}}(p) \leq R(p) \leq C_1(\log p)^3(\log\log p)^2$$

for sufficiently large $p \in \mathscr{P}_0$. Taking logarithms proves (3), since $\log\log\log p = o(\log\log p)$. $\square$

5. Analytic inputs and the mean estimate

5.1. Uniform linear-polynomial upper bounds

We use the following consequences of Henriot's corrected upper theorem [H, HE, New Theorem 5, p. 377]. For fixed $j \geq 1$, integer $1 \leq A \leq 16x^2$, and $x \geq 1$, with positive polynomial values,

$$\sum_{x<v\leq 2x} \tau_j(Av-1) \ll_j x\log(2x)^{j-1}, \tag{12}$$

$$\sum_{x<v\leq 2x} \mathbf{1}_{P^-(Av-1)>w} \ll \frac{xG(A)}{\log(2\min(w,x))}, \qquad G(A) = A/\varphi(A). \tag{13}$$

In the corrected theorem take degree 1, $\alpha = 1/2$, $\delta = 1/4$, and $\varepsilon = 1/1000 < 1/500$. The coefficient condition is uniform because $x \geq C_0(A+1)^{1/4}$ for all sufficiently large x in this range. For $p \nmid A$ the exact valuation-e density, $e \geq 1$, is $(p-1)/p^{e+1}$; for $p \mid A$ it is zero. Euler-product bounds give (12)–(13). The multiplicative-function growth constants can be chosen independently of the roughness cutoff. Bounded x has bounded A; the denominator in (13) stays bounded even if the value 1 is retained. Only the corrected upper theorem is used, not Henriot's lower theorem.

The other classical inputs are the dimension-one lower fundamental lemma of sieve theory, Mertens' estimates, the prime number theorem, and the standard harmonic divisor sums $\sum_{m \leq Z} \tau_j(m)/m \ll_j \log(2Z)^j$. The precise lower-sieve application is given next; [HR] is the general reference. The final submission must pin its exact theorem locator.

5.2. Lower mean

For $(u, 6) = 1$,

$$|S_{3u}| = \frac{3\tau(u^2) - 1}{2} \geq \tau(u^2) = \sum_{a \mid u} 2^{\omega(a)}.$$

Hence

$$\mu \geq \frac{1}{12} \sum_{\substack{am \leq Y \\ (am,6)=1 \\ P^-(12am-1) > w}} \frac{2^{\omega(a)}}{am}. \tag{14}$$

Restrict $a \leq Y^{2/3}$ and put $H = Y/a$. For $\sqrt{H} \leq T \leq H/2$ and squarefree e supported on primes at least 5 not dividing a, CRT gives

$$\#\{T < m \leq 2T : (m,6) = 1,\ e \mid 12am - 1\} = \frac{T}{3e} + O(1). \tag{15}$$

There are two classes modulo $6e$, giving an absolute error. Use lower-sieve weights of level $T^{1/4}$. Uniformly in these parameters, $\log(T^{1/4})/\log w \to \infty$, and the summed $O(1)$ remainders are $O(T^{1/4}) = o(T/\log w)$. The local forbidden density is $1/p$ for $p \geq 5$, $p \nmid a$, and zero at primes dividing a. The local product is therefore $\gg 1/\log w$, uniformly in a. The fundamental lemma gives $\gg T/\log w$ surviving m in each block.

Summing harmonically over these blocks gives $\gg \log H/\log w$. The standard estimate $\sum_{a \leq Z, (a,6)=1} 2^{\omega(a)}/a \asymp (\log Z)^2$, together with $\log(Y/a) \geq L/3$, proves $\mu \gg L^3/\log w$.

5.3. Upper mean and two weighted moments

In a dyadic box $r \asymp U, b \asymp B, c \asymp C$, choose a largest coordinate scale H_1. Fix the other coordinates; the linear coefficient is four times their product and is at most $16H_1^2$. Use (13), drop positive restrictions, and average $G(4ab) \leq 2G(a)G(b)$ with $\sum_{j \leq T} G(j) \ll T$. A nonempty rough box has $w < 32H_1^3$, so $\log(2\min(w, H_1)) \gg \log w$. Its reciprocal weighted contribution is $O(1/\log w)$. There are $O(L^3)$ boxes, proving the upper mean in (5).

The same largest-coordinate argument using (12) gives, for $j = 2, 3$,

$$\mathcal{S}_j := \sum_{(Q,s) \in \mathcal{A}_Y} \frac{\tau_j(Q)}{Q} \ll L^{j+2}, \tag{16}$$

$$\sum_{(Q,s)\in\mathcal{A}_Y} \frac{\tau(Q)}{Q}\frac{b}{c}\log(2Lc/b) \ll L^3\log(2L). \tag{17}$$

Indeed, (12) costs L^{j-1} per reciprocal weighted box. For (17), summing over the dyadic difference of the b, c scales uses $\sum_{h\geq 0} 2^{-h}(\log(2L)+h) \ll \log(2L)$; only two independent scale sums remain, alongside the single L from the divisor average. Boundary boxes with a coordinate 1 are included.

6. Positive divisor expansion and the tail

For a full divisor $d > 1$ of a retained modulus and a literal label $s \leq X$, write

$$H_{s,d} = \sum_{\substack{Q=dq \text{ retained} \\ s\in S_{(Q+1)/4}}} \frac{1}{dq}.$$

If this is nonempty, let q_0 be its **first retained** cofactor and set

$$B_{s,d} = \begin{cases} (dq_0)^{-1}, & q_0 = 1 \text{ or } w < q_0 < h_s/L, \\ 0, & \text{otherwise,} \end{cases} \qquad T_{s,d} = H_{s,d} - B_{s,d}.$$

Both are zero for an empty sum. Every eligible cofactor satisfies $dq \equiv -1 \pmod{4h_s}$ and $(dq+1)/4 > s$. The congruence is one residue class because $(d, 4h_s) = 1$. Proper retained cofactors exceed w. After q_0 their spacing is at least $4h_s$; a first term assigned to T has mass at most $L/(dh_s)$. Harmonic summation yields

$$0 \leq T_{s,d} \ll \frac{L}{d\kappa(s)}. \tag{18}$$

For $a \in (\mathbb{Z}/d\mathbb{Z})^\times$, set $b_{d,a} = \sum_{s\equiv a(d)} B_{s,d}$ and $t_{d,a} = \sum_{s\equiv a(d)} T_{s,d}$. Use the norm

$$\|v\|^2 = \sum_{d>1} \varphi(d) \sum_{a\in(\mathbb{Z}/d\mathbb{Z})^\times} v_{d,a}^2.$$

Expanding $\|b+t\|^2$ sums **ordered** packet pairs and every divisor $d > 1$ of $\gcd(Q, Q', s-t)$. Its coefficient for a packet pair is exactly

$$\frac{\gcd(Q, Q', s-t) - 1}{QQ'}, \tag{19}$$

with $\gcd(Q, Q', 0) = \gcd(Q, Q')$. If the pair contributes to (4), its two orientations contribute $2(g-1)/(QQ') \geq g/(QQ')$. All other terms are nonnegative. Therefore

$$\Delta \leq \|b+t\|^2. \tag{20}$$

Neither same-row terms nor partially compatible divisors are deleted.

The representation $s = rb^2$ gives $\sum_{s\leq X} \kappa(s)^{-2} \ll 1$ and $\sum_{s\leq X} \kappa(s)^{-1} \ll L^2$. Rough harmonic summation gives

$$\sum_{\substack{1<d\leq 4X \\ P^-(d)>w}} \frac{1}{d} \ll \frac{L}{\log w}.$$

For $0 < |s-t| \le X$, the positive full-prime-power Euler product gives

$$\sum_{\substack{d \| |s-t| \\ d>1,\ P^-(d)>w}} \frac{1}{d} \ll \frac{L}{w \log w};$$

there are $O(L/\log w)$ possible prime factors, each exceeding w. Using $\varphi(d) \le d$ in (18), equal literals contribute $O(L^3/\log w)$ and unequal literals $O(L^7/(w \log w))$. Thus

$$\|t\|^2 \ll \frac{L^3}{\log w} + \frac{L^7}{w \log w} = O(\mu). \tag{21}$$

7. Primitive rays and nested pairs

7.1. Ray counting with all multiples included

Consider an anchor $s = rb^2$, $M = rbc$, and a compatible partner $t = r'\beta^2$, $M' = r'\beta c'$. Let d divide both row moduli and assume $s \equiv t \pmod d$. Put

$$\gamma = \frac{\operatorname{lcm}(3, r'\beta)}{r'\beta} \in \{1, 3\}, \quad a = c'/\gamma, \quad \ell = (a, \beta), \quad a = \ell a_0, \quad \beta = \ell \beta_0, \quad D_0 = a_0 \beta_0.$$

Writing $Q' = d\nu$ gives

$$4\gamma s a_0 \equiv \beta_0 \pmod d, \qquad d\nu + 1 = 4\gamma D_0 r' \ell^2, \qquad \beta_0 < \gamma a_0, \quad (a_0, \beta_0) = 1. \tag{22}$$

All cancelled factors are units modulo d. For fixed primitive ray, γ, and ν, unique squarefree decomposition determines at most one pair (r', ℓ), hence one original label. Also $(d, 4\gamma D_0) = 1$. The number of occurrences with $V \le \nu < 2V$ is at most $V/(4\gamma D_0) + 1$. The endpoint is charged once per primitive ray, not once per multiple.

Distinct positive primitive vectors in this congruence lattice have a nonzero determinant divisible by d. In a rectangle $a_0 \asymp U, \beta_0 \asymp B$, slopes are separated by $\gg d/U^2$, so there are $O(1 + UB/d)$ vectors. Covering a product shell by $O(\log(2D))$ such rectangles gives $O(\log(2D)(D/d + 1))$ rays. Thus, if all products considered satisfy $D_0 \ge d_0 \ge 1$,

$$\#\{D_0 \le T\} \ll T \log(2T)/d + \log^2(2T),$$

$$\sum_{d_0 \le D_0 \le T} \frac{1}{D_0} \ll \frac{\log^2(2T)}{d} + \frac{\log(2d_0)}{d_0}. \tag{23}$$

The counts and sums in (23) range over primitive rays, not merely distinct integer values of their products. These bounds include unbalanced rectangles.

7.2. Complete nested mass

Let $\mathcal{N}$ be the sum of $1/Q'$ over compatible labels on retained rows $Q < Q'$ with $Q \mid Q'$. Then $Q' = Q\nu$ with $\nu > w$. The vector $(c, \gamma b)$ is parallel to the anchor. For a partner of unequal slope the determinant has absolute value at least Q; hence

$$D_0 \ge \min\left(\frac{Q^2}{4\gamma c^2}, \frac{Q}{2\gamma b}\right) \ge \tfrac{1}{2} \min((rb)^2, rc). \tag{24}$$

For example, one of $c\beta_0$ or $\gamma b a_0$ is at least $Q/2$; use $\beta_0 < \gamma a_0$ in the first case and $\beta_0 \geq 1$ in the second.

Set $d_0 = \max(1, \min((rb)^2, rc)/2)$. For $r' > R$, (22) gives $D_0 < QV/R$. Equations (22)–(24) bound the weighted mass in a cofactor block for fixed Q, s, γ by

$$\ll \frac{L^2}{Q^2} + \frac{\log(2d_0)}{Qd_0} + \frac{L}{QR} + \frac{L^2}{QV}. \tag{25}$$

The sum of its second term over labels is $O(L)$, by enlarging to the positive sums

$$\sum_{rbc\leq X,\ b<c} \frac{\log(2rb)}{r^3b^3c} = O(L), \qquad \sum_{rbc\leq X,\ b<c} \frac{\log(2rc)}{r^2bc^2} = O(1).$$

Summing the cofactor blocks, using $\sum_V 1/V \ll 1/w$ and $\sum_{Q,s} Q^{-2} \leq \mu/w$, proves

$$\mathcal{N}_{\text{nonparallel},\ r'>R} \ll L^2 + \mu(L^3/w + L^2/R). \tag{26}$$

Here is the remaining cover of the nested mass. Same-literal later rows satisfy $Q\nu \equiv -1 \pmod{4h_s}$ and $Q \equiv -1 \pmod{4h_s}$, so $\nu \equiv 1 \pmod{4h_s}$. The retained anchor is the first cofactor, $\nu = 1$; every later cofactor is at least $1 + 4h_s$. All same-literal later terms therefore lie in the tail and cost $O(L \sum_{rbc\leq X} 1/(r^2b^2c)) = O(L^2)$.

For unequal literals, $t = s + kQ > Q$. Put $W = 2^{\omega(Q)}$. The inequality $1/\kappa(t) \leq \sum_{d_1^2 u = t} 1/(d_1 u)$ and the bound of W unit quadratic roots modulo the full odd Q give, for dyadic D, U,

$$\#\{d_1 \asymp D, u \asymp U : d_1^2 u \equiv s \pmod{Q}\} \ll \min\{D(U/Q+1), WU(D/Q+1)\}.$$

After reciprocal weighting this is bounded by a constant times $W/Q + \min(1/U, W/D)$, with the important restriction $D^2U > Q/8$. Let $D_* = (QW)^{1/3}$. For $D \leq D_*$ the endpoint sum is $O(D^2/Q)$. For $D > D_*$, retain $U_{\min} = \max(1, Q/(8D^2))$. Then

$$\sum_{U\geq U_{\min}\ \text{dyadic}} \min(1/U, W/D) \ll \frac{W}{D}\left(1 + \log^+ \frac{8D^3}{QW}\right).$$

At $D \asymp 2^j D_*$ this is $O((W/D_*)2^{-j}(1+j))$, a summable series in both regimes of $U_{\min}$. No extra logarithm is introduced. Consequently

$$\sum_{\substack{Q<t\leq X \\ t\equiv s\ (Q)}} \frac{1}{\kappa(t)} \ll W^{2/3}Q^{-1/3} + L^2W/Q. \tag{27}$$

Weighted Cauchy–Schwarz, $|S_M| \leq \tau_3(M)$, $\tau_3(M)^2 \leq \tau_9(M)$ and $4^{\omega(Q)} \leq \tau_4(Q)$ give $\sum_{Q,s} W/Q \ll L^{13/2}$. Thus (18) and (27) bound all unequal-literal tails and soft first terms by

$$O(L^{15/2}w^{-1/3} + L^{19/2}/w). \tag{28}$$

For squarefree $r' \leq R$, a label $t = r'\beta^2 \equiv s \pmod{Q}$ has at most W unit root classes; nonunit r' contributes none. Each label has at most one selected first row, of weight at most $\min(1/(Qw), 1/(3r'\beta^2))$. Split at $U = \sqrt{Qw/r'} < Q$. Each class has at most one representative below U; the tail includes both its first term and its spacing-Q integral. Its mass is $O(1/(Qw))$. Summing gives

$$O((R/w)L^{13/2}). \tag{29}$$

For parallel pairs write $b = jB, c = jC$, $(B, C) = 1$. Necessarily $\beta = \lambda B$, $c' = \lambda C$, and

$$r'\lambda^2 = rj^2 + kQ, \qquad k \geq 1, \qquad \nu = 1 + 4kBC.$$

Squarefree decomposition is unique. Retaining the finite later-row cutoff and $\nu > w$ gives mass per anchor $O(1/(Qw) + L/(QBC))$. The total is $O(\mu/w + L^2)$, since $\sum_{rj^2BC\leq X} 1/(rj^2B^2C^2) = O(L)$.

Taking $R = L^{93}$ in (26), (28), and (29), and using $w = L^{100}$ and the elementary bound $\mu = O(L^3)$, proves

$$\mathcal{N} \ll L^2. \tag{30}$$

The decomposition uses the first *retained* row throughout. Unretained eligible rows add no mass, and all subsequent eligible rows obey the same spacing bound. The square case $r' = 1$ is included in (29).

8. Equal slopes and hard proper first cofactors

8.1. Equal slopes

There is also a unique representation $M = kxy, s = kx^2$ with $x < y$ and $(x, y) = 1$. Put $a = xy$ and $Q_k = 4ak - 1$. Then $(Q_k, Q_{k'}) = (Q_k, k' - k)$, and this divides $s_{k'} - s_k$. For $d \mid Q_k$, later rows divisible by d have $k' = k + jd$, $j \geq 1$, and their reciprocal sum is $O(L/(ad))$. Thus the equal-slope off-diagonal expanded mass for fixed x, y is at most

$$\frac{CL}{a^2} \sum_{k\leq X/a} \frac{\tau(4ak-1)}{k}.$$

Divisor pairing gives $\sum_{k\leq H} \tau(4ak - 1) \ll H\log(2aH) + \sqrt{aH}$. Partial summation bounds the displayed reciprocal sum by $O(L^2 + \sqrt{a})$. Both $\sum_{x<y}(xy)^{-2}$ and $\sum_{x<y}(xy)^{-3/2}$ converge. Including the diagonal, whose total is at most μ, the complete equal-slope contribution is $O(L^3)$.

8.2. Unequal slopes at a hard anchor

Now d is an arbitrary divisor in (19), not necessarily the full gcd. A hard proper first anchor has

$$Q = dq = 4rbc - 1, \qquad w < q < h_s/L \leq 3rb/L,$$

so $d > Lc$. Applying the determinant argument modulo d gives

$$D_0 \geq \min\left(\frac{d^2}{4\gamma c^2}, \frac{d}{2\gamma b}\right) > \min(L^2/12, Lc/(6b)).$$

The reciprocal-shell endpoint in (23) is therefore $O(F_L(b, c))$, where

$$F_L(b, c) = \frac{\log(2L)}{L^2} + \frac{b\log(2Lc/b)}{Lc}. \tag{31}$$

For any retained unequal-slope partner $Q' = d\nu$, $\nu > w$, $r' > R$, the weight is $\varphi(d)/(QQ') \le 1/(Q\nu)$. As in (25), a cofactor block contributes

$$\ll \frac{L^2}{Qd} + \frac{F_L(b,c)}{Q} + \frac{L}{QR} + \frac{L^2}{QV}. \tag{32}$$

This estimate does not require the partner to be first or hard, and allows $Q' = Q$. Summing cofactor blocks and divisors $d \mid Q$, using $d > w$ and (16)–(17), gives

$$\text{large-squarefree-part mass} \ll L^3 \log(2L) + L^6/R + L^7/w. \tag{33}$$

In particular its endpoint is $L \sum_{Q,s} \tau(Q) F_L(b,c)/Q$; the first part is $(\log(2L)/L)\mathcal{S}_2$ and the second is (17). There is no pointwise substitution $\tau(Q) \ll \log Q$.

For squarefree $r' \le R$, only first partners enter $\|b\|^2$. The label $r'\beta^2$ occupies at most $2^{\omega(d)}$ unit root classes modulo d and selects at most one B row. Its weight is at most $\min(1/(Qw), d/(3Qr'\beta^2))$. Split at $U = \sqrt{dw/r'} < d$. The first term and integral tail in each class have combined mass $O(1/(Qw))$. Since

$$\sum_{d|Q} 2^{\omega(d)} = \tau(Q^2) \le \tau_3(Q),$$

(16) yields

$$\text{small-squarefree-part first-partner mass} \ll RL^5/w. \tag{34}$$

This is not asserted for all small-squarefree-part later rows; their contribution is already assigned to the tail norm (21).

9. Completion of the proposed overlap proof

The positive expansion of $\|b\|^2$ has the following exhaustive cover.

1. If either selected cofactor is 1, then d equals that row modulus. The diagonal and both nested orientations cost at most $\mu + 2\mathcal{N}$.
2. Equal slopes cost $O(L^3)$ by Section 8.1.
3. Otherwise both selected cofactors are hard and proper, with unequal slopes. Split the partner's squarefree part at R and apply (33) or (34). Its first-row property in (34) follows from membership in B.

The cover may overlap, which only enlarges a positive upper bound. It includes same-row different-label pairs for proper d. The cutoff here can be chosen independently of the cutoff used to prove (30). Taking $R = L^{10}$ and $w = L^{100}$ gives

$$\|b\|^2 \ll \mu + \mathcal{N} + L^3 + L^3 \log(2L) + L^6/R + L^7/w + RL^5/w \ll L^3 \log(2L).$$

Combine this with (21) and $\|b+t\|^2 \le 2\|b\|^2 + 2\|t\|^2$ to obtain $\|b+t\|^2 \ll L^3 \log(2L)$. Equation (20) proves the second estimate in (5); Section 5 proves the first. This completes the proposed argument for Proposition 3 and hence Theorem 1. □

10. A separate proposed $O(\mu)$ strengthening

Status of this section. The following refinement comes from a separate internally audited argument. It has not received the same scope of completed external-model review as Sections 2–9, independent human specialist certification, or a complete Lean certificate. The completed Lean certificate for the weaker prime exceptional-set rate does not establish this sharper claim. Its inclusion records the strengthening without making Theorem 1 depend on it.

Proposition 5 (proposed stronger overlap). With exactly the family, first-retained-cofactor split and norm of Sections 2 and 6,

$$\|b\|^2 = O(\mu), \qquad \|t\|^2 = O(\mu), \qquad \Delta = O(\mu). \tag{35}$$

Corollary 6 (proposed sharper exception bound). Subject to Proposition 5, the right-hand side of (1), and the bound for (2), can be replaced by

$$C\frac{N \log\log N}{(\log N)^3}. \tag{36}$$

The lower threshold is unchanged. The relative prime exception bound is $O(\log\log N/(\log N)^2)$, and Corollary 2 is unchanged.

10.1. Retaining roughness in the divisor moments

For $j = 2, 3$, apply the corrected upper theorem [HE] with the same parameters as Section 5.1 to the nonnegative multiplicative function

$$f_{j,w}(m) = \tau_j(m)\mathbf{1}_{P^-(m)>w}, \qquad f_{j,w}(1) = 1.$$

Its growth constants are independent of w: it is multiplicative on coprime inputs, $\tau_j(p^e) \le j^e$, and $\tau_j(m) \ll_{j,\varepsilon} m^\varepsilon$. Zero values are allowed by the upper theorem; no positive lower-growth hypothesis is used.

For $p \nmid A$, upper-bound local factors after the sieve prefactor are $1 - 1/p$ for $p \le \min(w, x)$ and $(1-1/p)^{-(j-1)}$ for $w < p \le x$. For $p \mid A$ the factor is 1. The removed small-prime sieve factors cost at most $G(A) = A/\varphi(A)$; removed large-prime factors only decrease the upper bound. Thus the refinement of (12)–(13) is

$$\sum_{x<v\le 2x} f_{j,w}(Av-1) \ll_j \frac{xG(A)\log(2x)^{j-1}}{\log(2\min(w,x))^j}, \qquad 1 \le A \le 16x^2. \tag{37}$$

Bounded x is handled as in Section 5.1, including the retained value 1.

The coefficient correction must be averaged, not treated as a bounded pointwise constant. Writing μ_{Mob} for the Möbius function,

$$G(n) = \sum_{d\mid n} \frac{\mu_{\mathrm{Mob}}(d)^2}{\varphi(d)}, \qquad \sum_{n\le T} G(n) \le T\prod_p \left(1 + \frac{1}{p(p-1)}\right) \ll T.$$

Together with $G(4uv) \le 2G(u)G(v)$, the largest-coordinate dyadic argument of Section 5.3 now proves

$$\mathcal{S}_j \ll \frac{L^{j+2}}{(\log w)^j}, \quad j = 2, 3, \qquad \mathcal{J} := \sum_{Q,s} \frac{\tau(Q)}{Q}\frac{b}{c}\log(2Lc/b) \ll \frac{L^3\log(2L)}{(\log w)^2}. \tag{38}$$

Indeed every nonempty rough box has $w < 32H_1^3$, so its denominator in (37) is bounded below by a constant times $(\log w)^j$ even when $H_1 < w$. Averaging G in the two fixed coordinates then gives the same scale sums as before, with these extra denominators retained.

10.2. Equal slopes and the exceptional coefficient range

The equal-slope estimate of Section 8.1, retaining roughness on the earlier row, gives off-diagonal mass at most

$$CL \sum_{x<y,(x,y)=1} a^{-2} \sum_{\substack{k\le X/a \\ P^-(4ak-1)>w}} \frac{\tau(4ak-1)}{k}, \qquad a = xy. \tag{39}$$

Split into $k \ge a$ and $k < a$. In a dyadic block $k \asymp H$ meeting the first range, $A = 4a \le 8H \le 16H^2$. A retained term also forces $w < 16H^2$, so (37) with $j = 2$ applies with denominator comparable to $(\log w)^2$. Dividing by $k \asymp H$ and summing blocks bounds this inner sum by $O(G(4a)L^2/(\log w)^2)$. Since $\sum_{x<y} G(4xy)/(xy)^2$ converges, the contribution to (39) is $O(L^3/(\log w)^2)$.

For $k < a$, roughness forces $a > \sqrt{w/4}$. The elementary divisor bound $\tau(m) \ll m^{1/8}$ gives $\sum_{k<a} \tau(4ak-1)/k \ll a^{1/4}$. There are at most $\tau(a) \ll a^{1/8}$ slopes with product a. This part of (39) is

$$\ll L \sum_{a>\sqrt{w/4}} a^{-2+1/4+1/8} \ll Lw^{-5/16} \ll Lw^{-1/4}.$$

It is not fed to (37) outside that theorem's coefficient range. Adding the diagonal of mass at most μ, the full equal-slope contribution is

$$\ll \mu + \frac{L^3}{(\log w)^2} + Lw^{-1/4} = O(\mu). \tag{40}$$

10.3. Reusing the geometry with the sharper moments

The determinant, primitive-ray, first-partner and tail estimates of Sections 6–8 remain unchanged. The endpoint from (31)–(32) is now

$$E := L \sum_{Q,s} \frac{\tau(Q)}{Q} F_L(b,c) \ll \frac{\log(2L)}{L} \mathcal{S}_2 + \mathcal{J} \ll \frac{L^3 \log(2L)}{(\log w)^2} = O(\mu). \tag{41}$$

For the large-squarefree-part partners, (32) summed over divisors and cofactor blocks gives

$$\ll E + (L^2/R + L^3/w)\mathcal{S}_2 \ll E + \frac{L^6}{R(\log w)^2} + \frac{L^7}{w(\log w)^2}.$$

The small-squarefree-part **first** partners contribute $(R/w)\mathcal{S}_3 \ll RL^5/(w(\log w)^3)$. These are the same partner classes as in (33)–(34), with (38) substituted for (16)–(17).

Use the complete positive cover of Section 9, (30), (40) and (41). With $R = L^{10}$ and $w = L^{100}$,

$$\begin{aligned} \|b\|^2 \ll \mu + L^2 + \frac{L^3}{(\log w)^2} + Lw^{-1/4} + \frac{L^3 \log(2L)}{(\log w)^2} \\ + \frac{L^6}{R(\log w)^2} + \frac{L^7}{w(\log w)^2} + \frac{RL^5}{w(\log w)^3} = O(\mu). \end{aligned} \tag{42}$$

Equation (21) already gives $\|t\|^2 = O(\mu)$. Therefore (20) and the norm-square inequality used in Section 9 give $\Delta = O(\mu)$, completing the proposed proof of Proposition 5.

Finally (8) gives $v = O(\mu)$, so (10) costs $O(H/\mu + Y^2)$ instead of $O(H(\log L)^3/L^3 + Y^2)$. The same choice $Y = N^{1/3}$ and dyadic summation prove (36). The proposed improvement is a factor $(\log\log N)^2$ in the exception bound, not a higher typical exponent or a universal assertion. No full-period exponential no-hit theorem or exponential interval admission statement is needed or claimed by this strengthening. □

11. Verification, provenance, and limitations

The mathematics above is the basis of the claim, not agreement among reviewers. AI systems were used substantially in developing arguments, checking identities, exploring counterexamples, reviewing drafts and sources, and implementing partial Lean formalizations. The project has used OpenAI Codex models and Anthropic Claude models. Such assistance and internal cross-checking do not constitute independent human peer review. No AI system is listed as an author. Brian Akaka is the sole author and is responsible for the manuscript's public claims.

11.1. Scope of the Lean formalizations

There are two complementary formal artifacts. The earlier finite artifact checks a concrete witness-to-abundance implication for the Section 2 packet family and includes the composite quadratic-root and interval helpers described below. A separate analytic artifact now gives an end-to-end proof of the prime exceptional-set lower bound and its relative-density conclusion. The latter uses a genuine restricted subfamily of the Section 2 witnesses, not an abstract mean or overlap hypothesis.

In the end-to-end artifact, put $B = 2^n$. The selected parameters satisfy

$$K \le a < b \le B, \qquad (a,b) = 1, \qquad B^4 < u \le B^6,$$

and define

$$M = 3abu, \qquad s = 3a^2u, \qquad Q = 12abu - 1.$$

The actual roughness cutoff is $\lfloor (\log(B^8))^{100} \rfloor$. These are valid members of the broader family in Section 2. The formal proof establishes, eventually,

$$\mu(2^n) \ge \frac{n^3}{A\log(n+1)}, \qquad \Delta(2^n) \le Dn^3$$

for explicit positive constants A, D. These weaker sufficient estimates imply the original prime exceptional-set rate; no upper mean estimate, Henriot theorem, prime number theorem, or unproved analytic interface is assumed.

For natural Y, A, H, r, assume $\mu_Y > 0$, $12Y < A$ and $r \le \mu_Y/2$. Let $\mathcal{B}_{\mathrm{II}}(A,H;r)$ consist of primes in $[A, A+H)$ which do not admit r distinct strictly ordered Type-II solutions. The finite formal theorem establishes

$$|\mathcal{B}_{\mathrm{II}}(A,H;r)| \le 4\left[\frac{H(\mu_Y + 2\Delta_Y)}{\mu_Y^2} + (1+12Y)^2\right]. \tag{43}$$

In Lean, having at least r solutions is expressed by an injective map from $\mathrm{Fin}(r)$ to the type of strict solutions, with the Type-II conditions on every image. Thus (43) counts failure of distinct solutions, not failure of distinct parameter descriptions. Its declaration is `interval_prime_typeII_bad_card_le` in `Conditional.lean`.

The checked chain includes decoder integrality and strict order, prime Type-II conditions and injectivity, exact single/pair CRT probabilities, the factor-two passage from the unordered overlap to the variance budget, two-sided residue-count endpoint bounds, and the full endpoint term in (43). The theorem assumes no analytic bound on μ_Y or Δ_Y. The positivity assumption is explicitly satisfiable: a checked example has $\mu_1 > 0$. This example does not establish an admissible positive threshold r or an informative numerical instance of (43); eventual usefulness still depends on the analytic estimates.

Supporting checked lemmas include positive full-divisor expansion, primitive-ray determinant bounds, squarefree-scale uniqueness, at most two unit square roots of a unit modulo an odd prime power, and arithmetic-progression reciprocal sums. The two independently developed packet models and their mean/overlap definitions are also proved equal.

The composite quadratic-root step is now checked as well. Write $\rho(q,u) = \#\{x \in \mathbb{Z}/q\mathbb{Z} : x^2 = u\}$. For every positive q and integer target a, the formalization proves the exact product

$$\rho(q,a) = \prod_{p|q} \rho(p^{v_p(q)}, a).$$

Every local modulus is the full prime power, not its radical. For odd $q > 0$ and a unit u, the checked cardinality is either 0 or $2^{\omega(q)}$, and hence at most $2^{\omega(q)}$. Solubility is not assumed merely from unitness. The modulus $q = 1$ is included: its unique residue contributes one root and the prime-factor product is empty.

For natural A, H the checked interval corollary is

$$\#\{0 \le k < H : (A+k)^2 \equiv u \pmod{q}\} \le \rho(q,u)\big(\lfloor H/q \rfloor + 1\big).$$

It requires only $q > 0$; the odd-unit case replaces $\rho(q,u)$ by $2^{\omega(q)}$. Thus the endpoint allowance is per root class, not one for the entire congruence. The empty interval is separately checked to have zero hits. These declarations are in `CompositeQuadraticRoots.lean` and `IntervalQuadraticRoots.lean`; this elementary finite-interval fact is not the outstanding analytic interval theorem for ESC exceptions.

An earlier formal lower-sieve interface incorrectly quantified an arbitrary roughness cutoff. Its falsity is now proved in Lean, and it is not an assumption of (43). The replacement fixes the cutoff as a function of Y, explicitly quantifies all $Y \ge Y_0$ for a uniform threshold Y_0, and retains the ranges in Section 5.2; its analytic lower bound is still an unproved proposition. This corrects a formal-interface error, not a counterexample to the manuscript's restricted sieve application.

The end-to-end declaration is

```
PrimeAbundance.prime_abundance
  : PrimeAbundance.PrimeAbundanceClaim
```

It proves the quantitative prime exceptional-set bound at the rate in (2) and proves that the exceptional primes have relative density zero. Its solution multiplicity is an injection into strictly ordered Type-II solutions. The proof contains concrete derivations of its lower mean, upper overlap, finite transfer, global asymptotic comparison and elementary prime-count denominator.

The verified implementation uses Lean 4.27.0 and mathlib commit `a3a10db0e9d66acb ebf76c5e6a135066525ac900`. The fresh-source verifier compiled all 24 local source files in the final import closure into a new temporary object directory; the ordinary project build completed 7,909 jobs. Axiom audits of the final theorem, analytic bound, quantitative bound, relative-density theorem and finite transfer report only `propext`, `Classical.choice` and `Quot.sound`; no `sorryAx` or project-specific axiom is used. The source and verifier are under `formalization/abundance-2026-09-05/coordination/ external-6pro-analytic/prime-abundance-analytic-local/`. Running `bash verify.sh` there reproduces the guarded fresh-source check against the pinned mathlib cache. This is not a source rebuild of all mathlib dependencies.

Remaining formal boundary. The certificate proves the prime lower-bound and relative-density portion, but it does not formalize the all-integer bound in (1), Proposition 3 exactly as stated for the full Section 2 family, or the Elsholtz–Tao upper-bound argument used in Corollary 2. Thus the matching prime exponent three follows by combining the new certified lower side with the cited published upper side, rather than from one end-to-end Lean theorem. The specific derivation in Sections 5–9 and the stronger $O(\mu)$ estimate of Section 10 remain outside the completed certificate. The earlier root helpers likewise do not certify estimates (27)–(29) for the full family.

11.2. Remaining mathematical review

This draft claims a complete Lean proof only for the prime exceptional-set lower bound and relative-density statement just specified. It does not claim a Lean proof of every manuscript assertion, a verified asymptotic leading constant, an effective numerical onset, a new zero-solution exception record, or a solution for every prime. It also makes no claim to literature-wide priority.

The highest-priority external checks are the uniform import (12)–(13), the lower-sieve application (15), the full-prime-power and endpoint accounting in Section 7, and the completeness of the positive cover in Section 9. Section 10 additionally requires scrutiny of the rough divisor moment (37), the averaged coefficient correction and the exceptional equal-slope range; the review status of the shorter route does not certify these additional steps. A submission version should include a frozen source archive, exact build instructions for whatever formal components are offered, and an explicit list of any analytic premises left unformalized.

References

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[HE] K. Henriot, *Nair–Tenenbaum uniform with respect to the discriminant—ERRATUM*, Mathematical Proceedings of the Cambridge Philosophical Society **157** (2014), 375–377. Erratum. The corrected New Theorem 5 on p. 377 is the imported upper theorem.

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[EP] C. Elsholtz and S. Planitzer, *The number of solutions of the Erdős–Straus Equation and sums of k unit fractions.* arXiv:1805.02945. Cited for related solution-count questions, not as a premise of our lower bound.

[D] B. Dahan, *Sieve dimension and search depth for the Erdős–Straus conjecture, $n \equiv 1 \pmod{24}$*, 25 August 2026. arXiv:2608.24035v1, Sections 4.1–4.4. Cited as related work requiring a complete priority comparison, not as a premise of Proposition 3.